

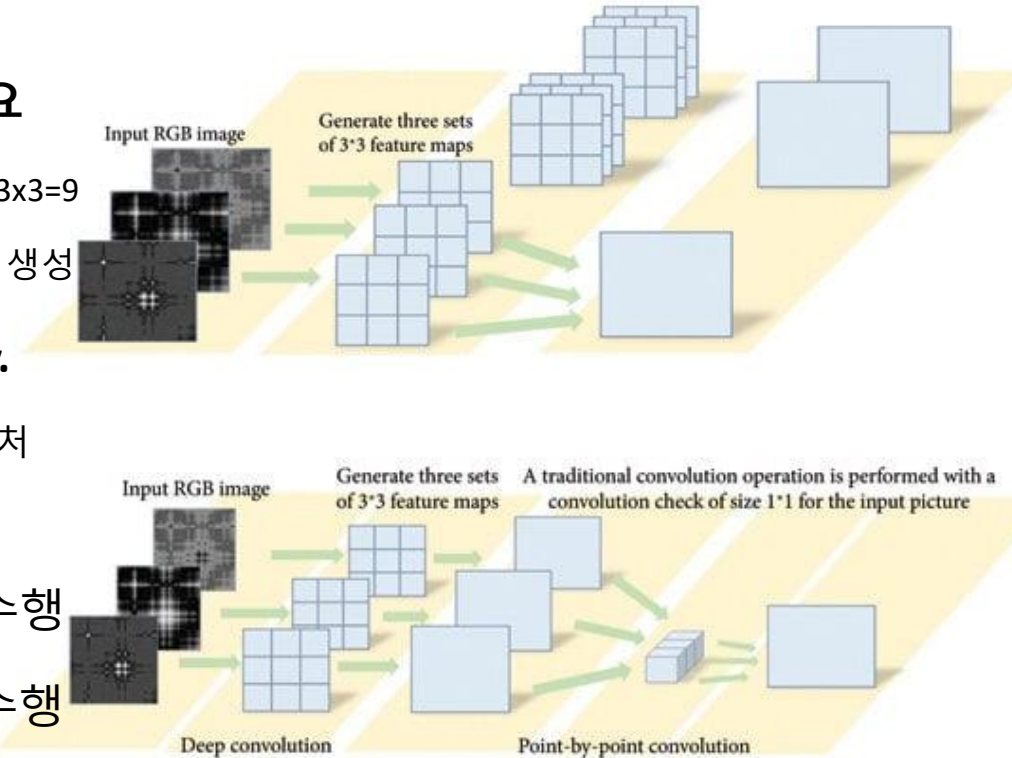
DS-CNN depthwise filter의 황금성배를 찾아서

Ocean's 8: The Quest for Universal Master Key Filters in DS-CNNs
(Babaiee et al., NerurIPS 2025)

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Background (1): CNN & DS-CNN

- **Conventional CNN: input channel x output channel 개의 kernel이 필요**
 - 우측 figur에서 input channel 이 RGB 3개, output channel 3개이므로, 필요 kernel 은 $3 \times 3 = 9$
 - R, G, B마다 각각의 kernel 이 적용된 후 합쳐져 하나의 feature map (output channel) 생성
- **Depthwise Separable CNN: Depthwise conv., and then Pointwise conv.**
 - (1) Depthwise: Input channel 마다 개별 kernel 적용해 공간(spatial) information을 캡처
 - (2) Pointwise: 1×1 conv 로 채널 정보를 섞어서 원하는 feature map을 만들
- Conventional CNN은 kernel 이 spatial & channel-wise comp. 를 모두 수행
- DS-CNN은 depthwise 가 spatial-wise를, pointwise가 channel-wise를 수행
- 따라서, **DS-CNN은 중복되는 공간 필터링이 줄어들어 매우 효율적**



(위) conventional CNN

(아래) Depthwise Separable CNN

Intelligent recognition for operation states of hydroelectric generating units based on data fusion and visualization analysis (International Journal of Intelligent Systems 2025)

Background (2): Patterns of Filters

- **DS-CNN의 depthwise filter에서 pattern이 발견되더라: filter를 Gaussian family 로 유형화 가능함을 발견**
- **Difference of Gaussian (DoG)**
 - Gaussian filter: 중심을 가장 크게 보고, 중심에서 멀어질수록 작게 보는 부드러운 필터
 - DoG = G_{small} (좁고 뾰족한 필터) – G_{large} (넓고 완만한 필터); 가까운 영역의 또렷함과 더 넓은 영역의 또렷함을 비교하는 필터
 - On-center DoG: 밝은 중심에 반응 / Off-center DoG: 어두운 중심에 반응
 - (DoG initialization) DS-CNN 의 depthwise filter에서 on/off-center DoG 패턴 발견 (WACV 2024)
 - (DoG-family initialization) 1M+ depthwise filter에서 DoG의 on/off-center, 1차 미분, 2차 미분 등 10개 패턴 유형화 (ICLR 2024)

(좌) on/off-center

Neural Echos: Depthwise Convolutional Filters
Replicate Biological Receptive Fields (WACV 2024)

(우) 유형화된 DoG family filters

UNVEILING THE UNSEEN: IDENTIFIABLE CLUSTERS
IN TRAINED DEPTHWISE CONVOLUTIONAL KERNELS
(ICLR 2024)

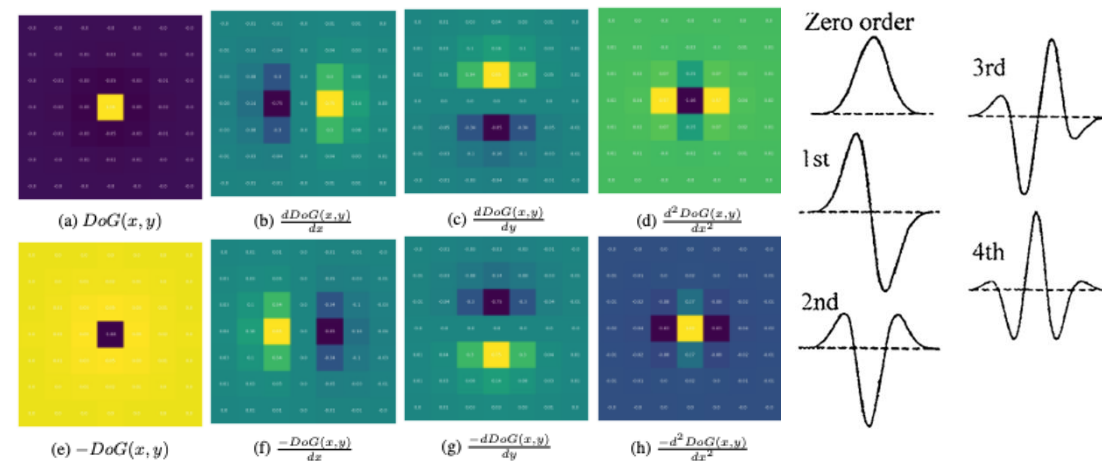
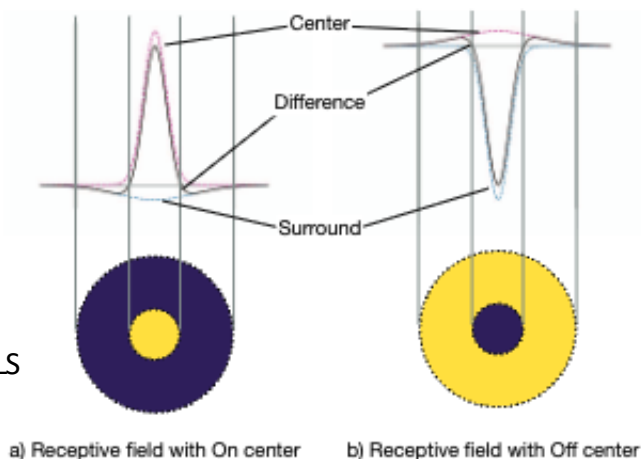


Figure 13: Complete diagram of the On and Off DoG filters and their derivatives on a 7x7 grid.

Background (3): Master Key Filters Hypothesis

- Depthwise filter를 유형화 가능하다면, master key filter가 존재할 수 있지 않을까?
- Conventional CNN의 통념: bottom layer는 선, 색, 모서리 feature를 캡처, top layer는 task-specific feature를 캡처
- Master Key Filters Hypothesis**
 - Dataset agnostic 하게 반복적으로 필요한 general spatial filter 그룹이 존재하며, DS-CNN의 depthwise filter는 dataset/task/architecture 가 달라도 해당 그룹으로 수렴하는 경향이 있다.
 - 단, 그룹에 몇 개 유형의 filter가 포함되는지는 모른다.
- Depthwise conv.를 freeze한 transfer learning 통해 **DS-CNN의 depthwise(spatial) filter는 task-specific 하지 않으며, dataset/architecture-agnostic 증명**

Depthwise-frozen transfer learning 설계 도식화
depthwise conv. Layer를 model A에서 model B로 가져와
freeze, different task & dataset 에 transfer learning

The Master Key Filters Hypothesis: Deep Filters Are General (AAAI 2025)



Figure 2: Overview of the experimental setup for depthwise filter transfers. Top: The base model-A is trained on the source dataset-A. Bottom: In the transfer model-B, the first n depthwise convolution layers of the network (in this example, $n = 3$) are transferred and frozen from the base model-A, the rest of the layers are randomly initialized, and then, they are trained on the target dataset-B. This experiment tests the extent to which the filters on layer n are general or specific.

Contribution points of this paper

- Master Key Filters Hypothesis를 minimal fundamental filter set으로 제한하였음
- DS-CNN의 depthwise conv. filters 가 8개 유형으로 축소 가능하며(basis of just 8 distinct filters), 나머지는 8개 유형의 선형변환에 불과함을 증명
- 8개 fundamental filters만 사용해도 performance integrity 를 유지함을 보여, 이 8개 filter set이 'essential visual processing primitive'임을 제시
- Background와 본 논문의 lineage

Paper	주장	근거 및 증명방식
Neural Echos (WACV 2024)	반복되는 패턴이 있다	DoG 패턴
Unveiling the Unseen (ICLR 2024)	패턴은 generalizable하다	1M+ filter를 clustering
Master Key Filters (AAAI 2025)	패턴에 해당하는 필터는 agnostic하다	Frozen-filter transfer
Ocean's 8 (NeurIPS 2025)	패턴에 해당하는 필터는 agnostic하 기 위해 8개면 충분하다	Greedy search & fixed-filter training

Methods: How to select 8 basis filters

- Pretrained DS-CNN 49개 모델 대상으로 모든 7x7 depthwise filter 수집 (1M+)
 - ConvNeXtV2, ConvNeXt, MogaNet, HorNet, ConvMixer: size 다르고, pretrained dataset이 다르며, pretrained dataset pixel size가 다름
 - Depthwise filter를 mean-centering & L2 normalization: filter의 offset 제거
- 1D autoencoder 학습: trained filter의 latent space 학습 (하기 figure2 참조)
 - Latent space in scalar $[0, 1]$ 로 형성되는데(by sigmoid), filter spectrum이 구성되면서 latent space에서 sampling하기에는 편리
 - AE의 latent space에서 50개 후보 filter 생성 (candidate filter sampling)
- original filter를 candidate filter의 affine transformation($aX+b$)으로 근사
 - thousands+ filter를 candidate filter로 대체하여, 작은 개수의 candidate filter로 유형화
- 정확도 기반 greedy search 진행
 - ConNeXtV2-Tiny 사용; reconstruction error 가 아닌 accuracy 기준
- 최종 8개 filter 선택 (post hoc)

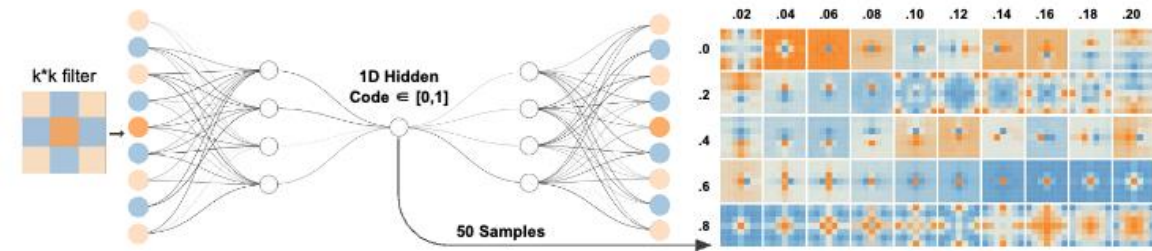


Figure 2: **Visualization of selecting candidate filters using autoencoder-based dimensionality reduction.** Left: An autoencoder compresses filters into a 1D hidden code. Right: Heatmap of 50 uniformly sampled candidate filters from 1D hidden code, generated by the decoder part of the autoencoder. These samples serve as the initial pool for our search for the master key filters.

Analysis: Understanding the 8 Filters

- 하단 figure 해석: 각 cell은 7x7 conv filter의 weight **주황색은 positive weight**, **파란색은 negative weight**
- **Filter 1, 2, 3, and 4: Central-difference-like patterns**
 - Center 및 그 주변에 강한(진한색) 값이 몰려 있음. 두 위치의 차이를 계산하는 edge detector
- **Filter 5, and 6: First derivatives of Gaussian**
 - Gaussian 1차 미분의 x/y 방향 형태와 유사. 각각 좌우 방향 밝기 변화와 상하 방향 밝기 변화를 의미하며, edge 방향/texture 감지
- **Filter 7: DoG (중심과 주변을 비교)**
 - Positive weight이 center에 강하고, 주변은 약한 negative weight. 중심과 주변의 대비를 계산하며, 점/contour 감지
- **Filter 8: Gaussian**
 - Center가 가장 높은 positive weight. 매우 작은 scale의 gaussian으로 해석될 수 있으며, smoothing/feature agg

(좌) 8 basis filter types
개인적으로는 7번은 부호반전된
gaussian 2nd derivative 아닌가 싶기도

(우) Gaussian 미분 그래프 모양

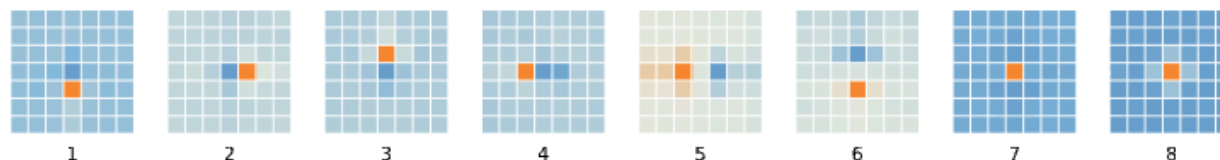
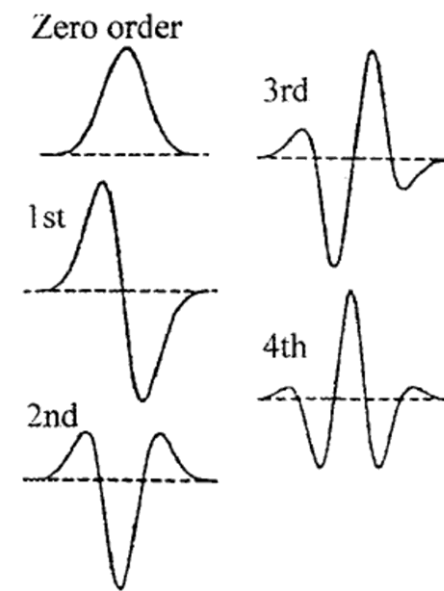


Figure 4: **Heatmap visualization of the eight universal filters discovered through systematic greedy search on the ConvNeXt2 tiny model.** Our empirical analysis demonstrates that DS-CNN filters predominantly converge to linear shifts ($ax+b$) of one of these eight filters, regardless of architecture or dataset. Filters 1-4 display central difference operator characteristics, and filters 5-8 correspond to established mathematical image processing fundamentals (See Figure 5).



Experiment (1): Do 8 filters preserve the pretrained model?

- 선택된 8개의 filter type이 실제로 효과적인가? (1) Pretrained Model 에 depthwise filter주입해서 실험
- Pretrained model의 depthwise kernel을 모두 8개 type의 affine variant로 변경해 w/o finetuning, ImageNet에 eval
- **8 filter types가 pretrained model에서 robust performance 를 증명**
 - ConvNeXtV2-Huge 50k filter 를 8개 type으로 affine transform 하여 사용해도 -0.5%p

– ConvNeXtV2-Huge

Table 1: Performance comparison when thousands of trained filters are replaced with linear shifts ($\mathbf{ax}+\mathbf{b}$) from candidate filters. *Without any fine-tuning* (For trained model see Table 2), models with just 8 selected filters from our greedy search on *ConvNeXtv2 Tiny* maintain remarkably high accuracy (e.g., only 3.5% drop for ConvNeXtv2 Huge despite reducing from 50k to 8 unique filter patterns) and even on different architecture, HorNet. Considering models' high sensitivity to filter alterations, this is evidence that DS-CNN filters predominantly converge to these filters.

ConvNeXtv2 Models	ConvNeXt					Hornet	
	Pico	Tiny	Base	Large	Huge	Tiny	Small
Number of Filters	2 944	6 624	18 048	27 072	49 632	11 488	17 232
Original Acc	80.3%	83.0%	84.9%	85.8%	86.3%	82.3%	83.5%
Acc with 50 candidates	75.0%	75.4%	80.5%	83.2%	84.0%	79.4%	81.3%
Acc with 25 candidates	72.0%	66.9%	72.8%	79.6%	80.4%	78.3%	80.9%
Acc with 10 candidates	23.4%	1.0%	1.4%	3.0%	2.0%	66.3%	70.5%
Acc with 8 (greedy search)	73.1%	76.7%	79.3%	81.2%	82.8%	76.0%	78.1%
Acc with 8 random filters	0.11%	0.10%	0.10%	0.12%	0.09%	0.96%	1.0%

Experiment (2): Training models with 8 filters

- 선택된 8개의 filter type이 실제로 효과적인가? (2) 8 type filters 를 freeze 하고 model을 from the scratch 학습
- **8 filter types가 from the scratch 모델 학습에서 original performance와 대등**
 - Original model과 8 filter types를 freeze한 from the scratch 학습 모델의 performance gap은 $[-0.5, +0.9]$ 에 불과 (좌측 table 2)
- 선택된 8개의 filter type이 실제로 효과적인가? (3) 위와 같은 환경에서 small dataset으로 학습한 경우
- **8 filter types가 small dataset에서도 positive inductive bias 를 보여 performance gain 이 있음**

Table 2: **ImageNet Top-1 accuracy comparison between conventional trainings and our 8-filter constraint.** Models restricted to using only our 8 unique filters (plus learnable bias terms) achieve comparable accuracy to their fully-trained counterparts. The consistent performance across different architectures (ConvNeXtv2 and HorNet) demonstrates the universality of these fundamental filters.

Models	ConvNeXtv2				HorNet
	Pico	Tiny	Base	Large	Tiny
Number of Original Filters	2 944	6 624	18 048	27 072	11 488
Original model with FCMAE ¹	80.3%	82.9% ²	84.9%	85.8%	—
Original model	79.7%	82.5%	84.3%	84.5%	82.3%
Our 8 unique filters + bias	80.2%	82.7%	84.6%	85.4%	81.8%

¹ FCMAE (Fully Convolutional Masked Autoencoder Framework) is a heavy pretraining.

² This accuracy is from the model released by the ConvNeXtV2's official paper repository. We trained the model using their exact same script and training parameters (with the default random seed set to 0) and achieved an accuracy of 82.7%, matching the result obtained with our 8-filter configuration.

Table 3: **Cross-dataset evaluation** demonstrating the superiority of our universal filter approach on smaller datasets. The ConvNeXt Femto model restricted to using only 8 unique frozen filters outperforms both models trained from scratch and those using transferred ImageNet filters, on Oxford Pets and Oxford Flowers. We evaluate multiple model sizes (Atto, Femto, Pico, Tiny) for Flowers and Pets datasets to verify that our observed advantage is consistent across architectures of varying capacity and not merely an artifact of dataset size limitations.

Dataset	CIFAR10	STL-10	Oxford Flowers				Oxford Pets			
			2040				3680			
# Training Set Size	50000	5000								
Model	Femto	Femto	Atto	Femto	Pico	Tiny	Atto	Femto	Pico	Tiny
Filters										
Original (normal training)	96.9	80.4	63.3	66.0	60.2	75.7	38.4	36.3	40.1	65.4
ImageNet Transferring	97.1	83.2	72.2	73.2	74.8	81.8	58.5	56.0	66.3	80.1
Our 8 Unique Filters	96.3	83.1	77.8	77.7	77.2	85.1	66.5	66.4	72.8	81.8

Conclusion & Limitation

- **Conclusion**

- 49 pretrained DS-CNN models 의 1M+ 7x7 depthwise filter를 8 type template으로 유형화
- 8 type filters를 사용한 Post-hoc replacement에서 정확도가 소폭 감소하나 robust 하고, from the scratch 학습에서는 baseline model과 비슷한 성능을 보이며, small dataset에서 fixed 8 type filters가 inductive bias로 긍정적인 작용

- **Limitation**

- 8 type templates는 ConvNeXtV2-Tiny 기반의 greedy search로 선택되었으며, elbow가 엄밀하지 않음 (figure 3)
- 8 type templates를 사용한 학습의 경우, pointwise conv가 학습 type templates 덕분이라고 보기 힘들
- Experiment가 7x7 kernel에 한정
- **결론: 제목대로 'Universal Master Key Filters'라고 할 수 있는**

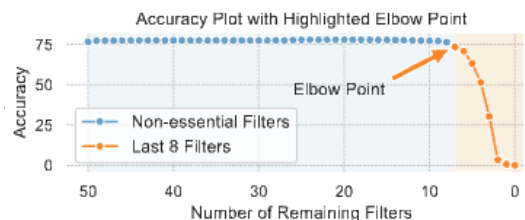


Figure 3: **Our systematic greedy search for the essential filters A.1.** While the removal of most filters did not noticeably change accuracy, 8 of them were essential, consistently in all models we tested.

Thank You!

Q&A